

CONEPRO · DESIGN DECISIONS REGISTER

ConePro design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/conepro		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open items from the review note, the decision records and the build plan work

ConePro design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note `docs/REVIEW.md` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

Table 1. Decisions awaiting Amish.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction: M12 rod joints and a 12 mm cone shoulder, pinned stop collar and welded handle, magnet pockets outside the strike area, one-piece clamp against the anvil, curved pad under the band, reel internals with the drum's edge under the exit, fixings for the reel and logger, reel-to-logger cable, coupling sleeve on the extension rod (P1 to P12)	Accept all; accept some and ask for alternatives to others	Accept all; each keeps what ConePro does and was needed to make a part buildable	Every component; the whole build plan follows these changes	CNP-DDR-004, Tables 1 and 2
2	Keeping the hammer on the upper rod when it is off the anvil (packing and carrying): the 8 kg hammer can slide off the rod's lower end	(a) a screw-on end cap on the rod's M12 stud whenever it is off the anvil; (b) pack with the anvil on the upper rod and relax the 10 kg heaviest-piece limit of R10 (about 11.3 kg); (c) user guidance only	(a); it adds to the safety case, so it is Amish's call	A small cap in BOM line 14; packing	CNP-DDR-004, A1
3	First co-design partner and user group	Small contractors, an NGO shelter or water team, or a low-volume road agency	None yet; to be picked per area	Not part of the TRL 3 build; shapes field trials at TRL 4	CNP-DDR-001 and CNP-DDR-002, O1
4	Clear windows in the reel housing and logger lid shown in the photoreal renders	(a) render-only features, opaque parts in the design; (b) real inspection windows, each with an extra seal	(a), since each window adds a seal to an IP65 box	None if (a); housing and lid if (b)	docs/REVIEW.md , 2026-09-26, item 2
5	Grip grooves on the hammer in the appearance model	Accept; leave the hammer plain	Accept; length is set at machining to keep 8.00 kg (the magnet pockets are now defined in CNP-DDR-004)	Hammer making sketch (grooves would be added)	docs/REVIEW.md , 2026-09-26, item 3
6	Rubber grips on the T-handle	Add grips of about 33 mm diameter to BOM line 6; no grips	Add them (a few dollars)	T-handle	docs/REVIEW.md , 2026-09-26, item 4
7	Coiled cable path as drawn in the renders	Accept; redraw	Accept	None	docs/REVIEW.md , 2026-09-26, item 5

■ To confirm when parts are bought

Table 2. Items to check against the parts actually bought.

#	What to confirm	Why it matters	Source
1	The tolerances of ASTM D6951 for the cone, hammer mass and drop, against a copy of the standard	R1 is met only nominally until the tolerances are checked	CNP-CAL-001 section A
2	The bubble level is 20 mm across and about 6 mm tall	It sits in a 20 x 3 mm recess in the seat disc	CNP-DDR-004, P4
3	The spring motor fits in 40 mm diameter x 8 mm and gives about 3 N retracted to 5 N extended	The reel layout and the wire slack and snap calculations assume it	CNP-CAL-001 [E5]
4	The angle sensor board is about 30 x 30 mm, 12-bit, with a matching diametric magnet	It mounts on the lid 1.4 mm from the shaft magnet	CNP-DDR-004, P8
5	The flanged IP65 logger box, about 100 x 70 x 42 mm, and its flange hole spacing (82 mm across, 36 mm along)	It sets the four tapped holes in the plate	CNP-DDR-004, P9
6	The M12 glands seal on 5 mm and 6 mm cable	IP65 entries for both cables (R11)	CNP-DDR-004, P10, P11
7	The band clamp's range takes the anvil plus the pad (about 90 mm across)	It goes round the anvil and over the pad	CNP-DDR-004, P7
8	The Hall switch and accelerometer are rated to 10,000 g shock, and the elastomer gives about 300 Hz isolation with the pad	The shock calculation assumes both	CNP-CAL-001 section G
9	The bearings are 6 x 13 x 5 mm and fit the 18 mm boss	The boss is printed to suit	CNP-DDR-004, P8

■ Value engineering

Value-engineering target: USD 400 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 335 for the instrument (USD 65 under the target); the kit with both optional accessories is USD 401 (USD 1 over the target).

- **Main cost drivers:** the mechanical penetrometer (items 1 to 7, USD 174), sensing and logging (items 8 to 12 and 17, USD 113), and the carry bag, hardware and consumables (USD 48); the optional lever (USD 50) and extension rod (USD 16) are outside the instrument total. The design for construction added USD 16 (the reel-to-logger cable at USD 6 and USD 10 of welding, fixings and reel parts).
- **Savings worth trying:** trim the price of an optional accessory, which would bring the full kit to the target; and re-price the sensing parts at purchase.

■ Decisions made

Table 3. Decisions made, with the record that holds each.

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D8: accelerometer tilt with a bubble level backup; optional lever extraction; sensor set designed to fit standard DCPs; draw-wire depth with a laser fallback; standard 8 kg geometry; Hall blow detection with cross-checks; electronics on the plate; three AA cells	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	CNP-DDR-001
2026-09-25	D9 lighter carried set (6 mm plate, aluminum clamp, 0.40 kg bag); D10 R5 at 850 mm per rod with an optional extension rod; D11 reel details and 850 mm usable range; D12 disposable-cone practice for stiff ground	Amish: "i accept all your recommendations, go with them across all repos."	CNP-DDR-002
2026-09-27	Keep the 60 mm drum and grow the reel housing to 72 x 60 x 72 mm	Amish asked to "resolve the challenges for ConePro, BridgePulse, Grainguard and WellSense" and chose this option	CNP-DDR-003
2026-09-30	Make the design physically buildable while drawing the build plan; changes recorded in CNP-DDR-004 and open for review (open decision 1)	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	CNP-DDR-004
2026-09-30	Open decisions go in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	This register